

Ke Li

Personal Website: <https://kelichloe.github.io>

Research Interests

My research lies at the intersection of AI/ML, optimization, and decision making under uncertainty. I pursue two complementary streams:

- **Data-Driven Optimization:** developing data-driven machine learning algorithms to support robust decision-making under uncertainty.
- **AI for Science/Human-AI Collaboration:** studying how human-AI collaboration can augment the scientific process, particularly in theory building.

Education

INSEAD	France
PhD Candidate in Decision Sciences (AI/ML + Optimization)	09/2023 - Ongoing
Advisor: Spyros Zoumpoulis	
The Chinese University of Hong Kong, Shenzhen	China
B.Eng (Honors), Computer Science and Engineering	09/2018 - 06/2022
China National College Entrance Examination (Gaokao)	China
Ranked in the Top 0.1% (among ~1 million candidates) in Henan Province	2018

Honors & Awards

Teaching Excellence Award, INSEAD	2025
“Stars of Tomorrow” Award of Excellent Intern, Microsoft Research Asia	2022
ExploreCSR Computing Research Award, Google Research & UTRGV [link]	2021
Undergraduate Research Awards, CUHK-Shenzhen	2020
Undergraduate Student Teaching Fellows, CUHK-Shenzhen	2020
China National Encouragement Scholarship, Ministry of Education of China	2019
Dean’s List, CUHK-Shenzhen	2019-20
Admission Fellowship of CNY 90,000/year, CUHK-Shenzhen	2018-22

Working Papers

- **Decision-Aware Segmentation** - Joint work with [Sandeep Chandukala](#), [Ernst Osinga](#), and [Spyros Zoumpoulis](#).

Presentations (forthcoming):

- INFORMS Annual Meeting, San Francisco, 2026
- INFORMS M&SOM Conference, 2026
- INFORMS Society for Marketing Science (ISMS), Lisbon, 2026
- Theory + Practice in Marketing Conference (TPM), Esade, Barcelona, 2026
- Singapore Rising Scholars Conference, SMU, Singapore, 2026

- **A Human-AI Collaborative Framework for Theory Building** - Joint work with [Philip Parker](#), [Phanish Puranam](#), [Eric Uhlmann](#), and [Spyros Zoumpoulis](#).

Presentations:

- INFORMS Annual Meeting, Atlanta, 2025
- AI Plus Management Doctoral Consortium, London, 2025

Publications - [See Google Scholar](#)

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1. Zhou, B., Li, K., Jiang, J., & Lu, Z. Learning from Visual Observation via Offline Pretrained State-to-Go Transformer. *Advances in Neural Information Processing Systems (NeurIPS, Top-tier AI/CS/ML conference)*, 2023.
 2. Dong, J., Li, K., Li, S., & Wang, B. Combinatorial bandits under strategic manipulations. In *the Fifteenth ACM International Conference on Web Search and Data Mining (WSDM, Top-tier data mining conference)*, 2022.
 3. Liu, Y., Li, K., Huang, Z., Li, B., Wang, G., & Cai, W. EduChain: a blockchain-based education data management system. In *Blockchain Technology and Application: Third CCF China Blockchain Conference*, 2021.

Industry Experiences

Beijing Academy of Artificial Intelligence

Reinforcement Learning Researcher (Full-time)

Beijing

12/2022 - 07/2023

- Designed and trained intelligent agents capable of learning directly from high-dimensional visual observations in complex game environments such as Minecraft.
- Developed novel architectures combining convolutional encoders with deep reinforcement learning algorithms (e.g., PPO, DQN)
- Led extensive experiments to validate model performance, and collaborated closely with researchers from Peking University. Our research paper ([website](#)) is accepted by NeurIPS 2023.

Microsoft Research Asia

Research Intern, Award of Excellent Intern, Innovation Engineering Group

Beijing

01/2022 - 06/2022

- Investigated model compression techniques for Transformer/BERT models, such as pruning, quantization, and distillation.
- Implemented compression pipelines and reproduced results from recent research papers.
- Optimized Transformer models for deployment by reducing parameters and memory footprint.
- Collaborated with researchers to refine experimental settings and improve reproducibility.

SenseTime

Reinforcement Learning Intern, OpenDI Lab

Shenzhen

08/2021 - 12/2021

ByteDance Technology

Machine Learning Intern, Data AI Lab

Beijing

12/2020 - 05/2021

Contributions to Open-Source Community

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- **Microsoft Research Asia** — Contributed to Microsoft's AutoML toolkit [NNI](#) (14,300+ GitHub stars), including transformer model-compression modules now integrated into both internal and external AutoML pipelines.

- **SenseTime (OpenDI Lab)** — Contributed 1,600+ lines of code to the [DI-engine](#) reinforcement-learning framework (3,600+ GitHub stars), improving algorithmic stability and large-scale training utilities.

Teaching Experience

INSEAD, Teaching Assistant

Winner of Teaching Excellence Award

- Foundations of AI for Managers, MBA Course 2025.05 - 2025.06
- Foundations of AI for Managers, MBA Course 2025.01 - 2025.02
- Probability and Statistics, PhD Course 2024.10 - 2024.12

CUHK-Shenzhen, Undergraduate Teaching Assistant

MAT1010 Calculus I

Fall 2019-2020

Academic Services

Session Chair

- INFORMS Annual Meeting (co-chair with Professor Spyros Zoumpoulis) 2025
Session Title: Estimation and Optimization in Data-Driven Decisions

Reviewer

- NeurIPS (Top-tier conference in CS/ML/AI) 2024

Beyond Academia

I am a part-time model in Paris ([\[portfolio\]](#)).

Patents

Personnel exemption aided decision making method and system based on historical big data. China Patent CN113673943A. [\[link\]](#)

Miscellaneous

Alumni Interview Artical ([EN](#) version/[CN](#) version)